

Mathematics A

Unit 1 • Chapter OL-T57 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 11 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T57 • Expertise Q16+ • 3 items • 11 marks

Chapter	Items	Marks	Starts
01. OL-T57 Probability Diagrams - Venn & Tree Diagrams	3	11	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Probability Diagrams - Venn & Tree Diagrams

Unit 1 • Expertise • 11 marks • 3 item(s)

June 2024 | 4MA1-2H Q16 | 7 marks

Read regions and probabilities from a Venn diagram

November 2021 | 4MA1-1H Q16 | 2 marks

Solve unknown Venn regions using a total

November 2021 | 4MA1-1H Q16 | 2 marks

Complete a three-set Venn diagram

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q16

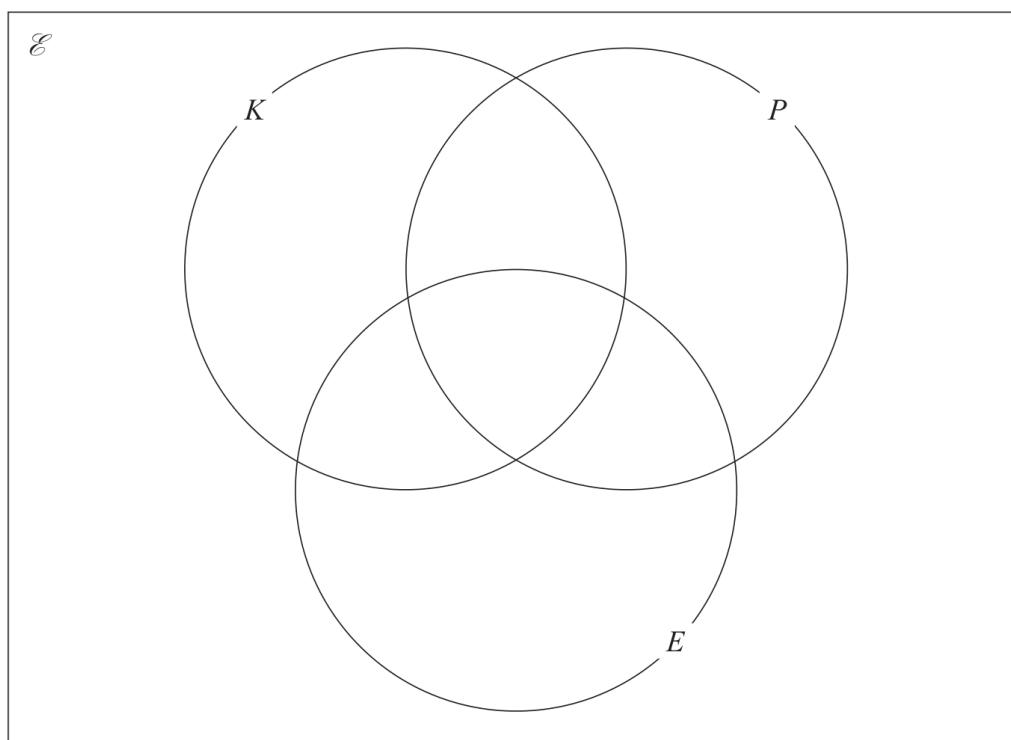
7 marks

- 16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

- (a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



(3)

4MA1-2H Q16

7 marks

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

.....
(2)

(c) Find $n(P \cap K')$

.....
(1)

(d) Find $n([P \cup E] \cap K)$

.....
(1)

4MA1-2H Q16

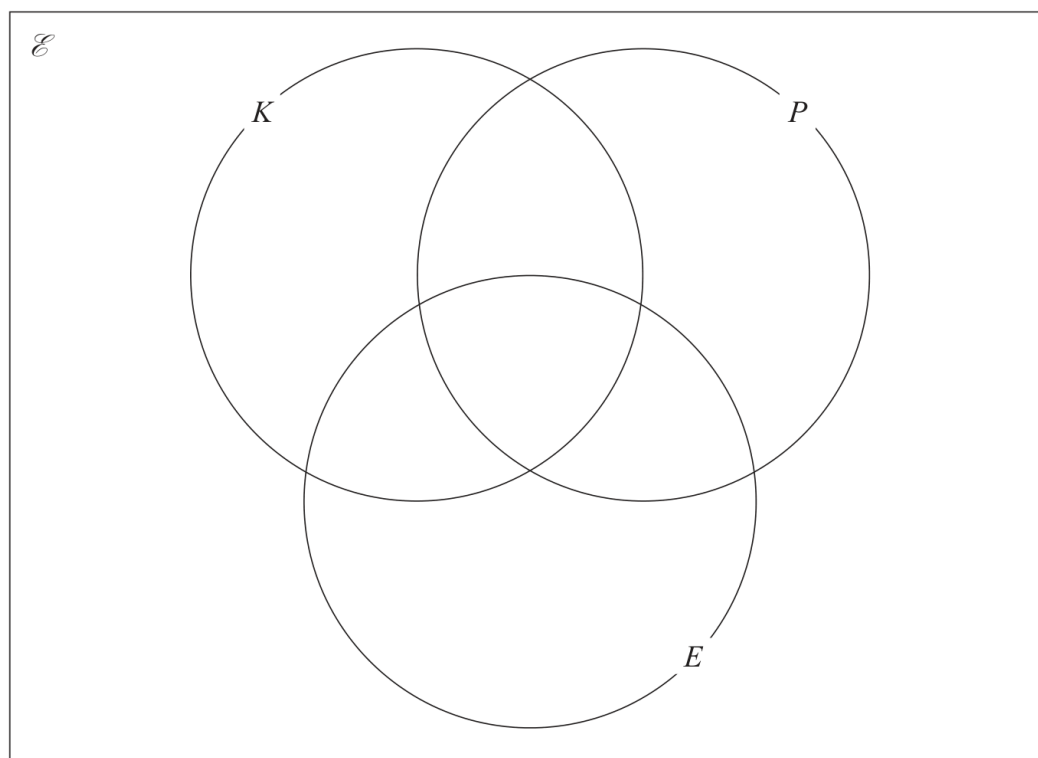
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(1)

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(1)

Three-set Venn diagram

7 marks

Regions

$$KPE = 9, KP = 8, PE = 7, KE = 11, P = 4, E = 12, K = 7, \text{ outside} = 2.$$

Conditional probability

$$P(K | P) = 17/28.$$

Counts

$$n(P \cap K') = 4 + 7 = 11; \quad n((P \cup E) \cap K) = 8 + 11 + 9 = 28.$$

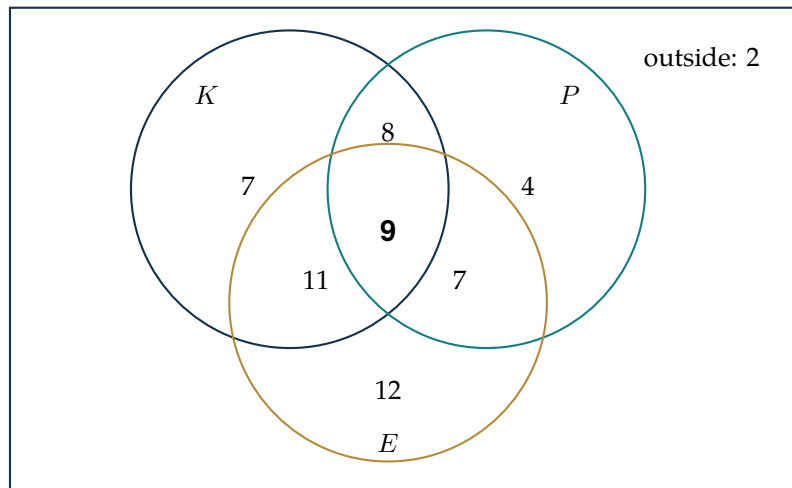
Final answer

$$\frac{17}{28}; \quad 11; \quad 28$$

Three-set Venn diagram

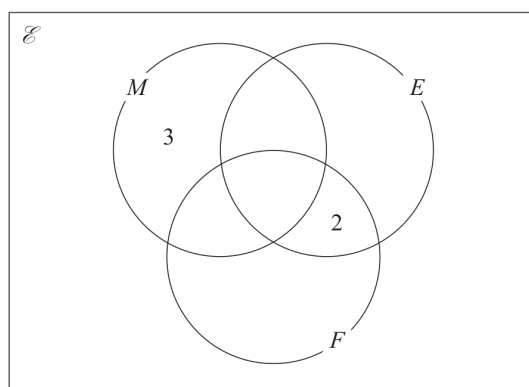
7 marks

Completed solution diagram: Completed three-set Venn diagram



- 16** There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

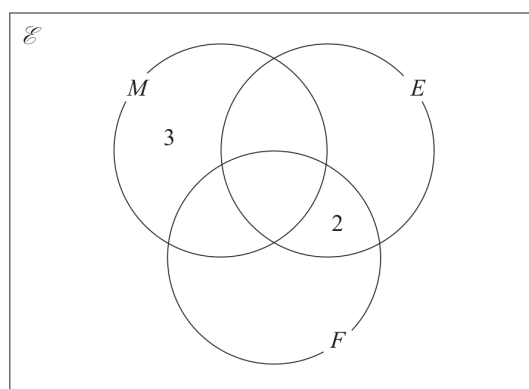
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$x = \dots\dots\dots$
 (2)

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$\dots\dots\dots$
 (1)

Worked solution

November 2021 | Question 16 | 2 marks

Family: Represent and read events with Venn diagrams • Variant: Solve unknown Venn regions using a total

Method / working

1. Use the Maths circle.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $M \text{ only} = 3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $M \cap E \text{ only} = 18 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $M \cap F \text{ only} = 11 - x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The total in $\setminus(M)$ is $\setminus(25)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $3 + (18 - x) + (11 - x) + x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $32 - x = 25$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $x = 7$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. So:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $M \cap E \text{ only} = 11$, $M \cap F \text{ only} = 4$, $E \cap F \text{ only} = 2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. $E \text{ only} = 20 - 11 - 2 - 7 = 0$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $F \text{ only} = 14 - 4 - 2 - 7 = 1$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. For a student who passed Maths, failed French means $\setminus(M)$ but not $\setminus(F)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $\frac{3+11}{25} = \frac{14}{25}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

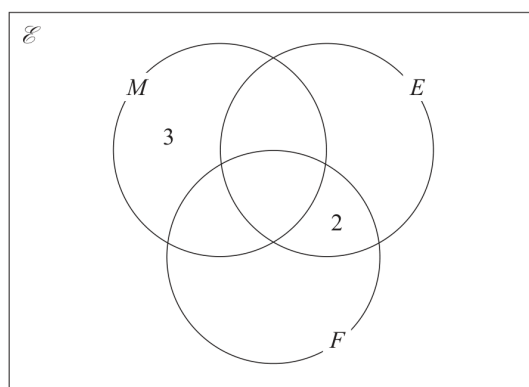
15. Answers: $\setminus(x=7)$, and $\setminus(\frac{14}{25})$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $x=7$, and $\frac{14}{25}$.

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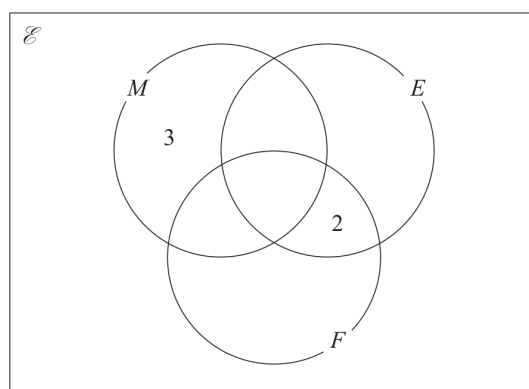
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Final answer: $x=7$, and $\frac{14}{25}$.